

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

May 2016

EC 303/EC 303.01 Audio Video Engineering

Date: 07.05.2016, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

<u>SECTION - I</u>		
Q 1	Define following terms.	[07]
(a)	SRI	
(b)	Sabine's Formula	
(c)	Intervals	
(d)	Octaves	
(e)	Overtones	
(f)	Timbre	
(g)	Harmonics	
Q - 2 (a)	Draw and Explain basic PA System.	[05]
(b)	Explain working principle of a Moving Coil cone type loudspeaker.	[04]
(c)	Define reverberation time. Explain the importance of reverberation	[05]
<u>OR</u>		
Q - 2 (a)	Explain the Condenser type microphone.	[06]
(b)	Explain Horn Type Loudspeaker.	[04]
(c)	Explain growth and decay of sound in an Enclosure.	[04]
Q - 3	Write Short note on followings (Any Two):	[14]
(a)	3D Television	
(b)	Hypermedia and Multimedia	
(c)	HDTV	
(d)	Flat panel Display TV Receiver	
<u>SECTION – II</u>		
Q - 4 (a)	Draw block diagram of a monochrome TV Transmitter	[04]
(b)	Define Aspect ratio with its importance	[03]

Q - 5 (a)	What do you understand by interlaced scanning?	[05]
(b)	Explain Functions of vertical pulse train	[04]
(c)	Explain merits & Demerits of negative modulation	[05]
	<u>OR</u>	
Q - 5 (a)	Explain Vestigial Sideband Transmission	[05]
(b)	What is Adjacent Channel interference?	[04]
(c)	Write a short note on Interference suffered by TV signal	[05]
Q - 6 (a)	Sketch the horizontal line and sync details. Also explain importance of Front Porch and Back Porch.	[06]
(b)	What is Image Rejection Ratio?	[04]
(c)	Explain SAW Filter	[04]
	<u>OR</u>	
Q – 6 (a)	Explain VHF Tuner	[06]
(b)	Explain the Space wave propagation	[04]
(c)	What is AGC?	[04]
